

# ROBOTICS (05/09/2016)

Matteo Matteucci, Gianluca Bardaro

The exam will be graded IFF the following recommendations have been taken into account:

- Write clearly so that the teacher can easily understand your answers
- Write your name, surname, and student id on each sheet you deliver for evaluation
- For each exercise/question report clearly the number and sub-number (if present)
- You are not allowed to use any programmable device (e.g., smartphone, calculator, etc.)
- You can use pen or pencil, paper will be provided, you cannot use notes or books

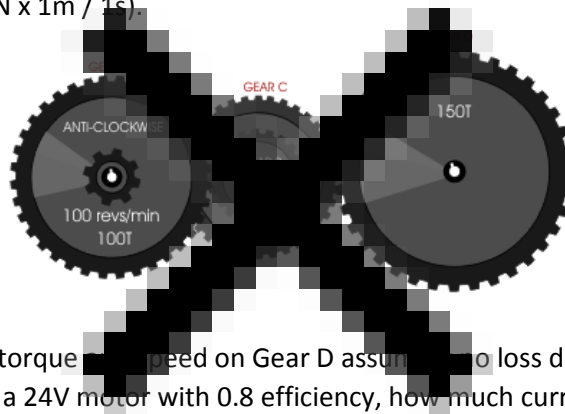
## **Exercise 1 (Algorithm)**

Let's consider Simultaneous Localization and Mapping, in particular:

- a) Describe the Simultaneous Localization and Mapping problem
- b) Describe the difference between online SLAM and full SLAM
- c) Describe what a particle filter is and how it can be used for SLAM or localization
- d) Describe the FastSLAM algorithm for full SLAM

## **Exercise 2 (Numerical Exercise)**

Let's consider the group of gears in the picture where Gear A spins at 100 revolutions per minute driven by a 100W motor (recall  $1W = 1N \times 1m / 1s$ )



- a) What are the output torque and speed on Gear D assuming no loss due to friction?
- b) Assume we are using a 24V motor with 0.8 efficiency, how much current is it driving?

## **Exercise 3 (Theory)**

Describe the following kinematics and provide an example for each of them:

- a) Differential drive
- b) Synchronous drive
- c) Tricycle kinematics
- d) Ackerman steering

## **Exercise 4 (ROS)**

What are nodes and messages in ROS and how are they used to develop a robot? Make an example of a simple robot and its possible ROS architecture.

## **Exercise 5 (Other)**

Describe the Beam Sensor Model, its components, and the rationale behind each of them.